

Scandinavian flight 751

Flight model: Douglas DC-9
Date: December 27, 1991

From: Stockholm, Sweden
To: Warsaw, Poland

The weather in Stockholm was snowy. The tarmac is covered with snow. A DC-9 has a 2 rear engines. These rear engines are mounted on back of the wings.

Problem

Right after take off, The crew hear some unusual sound. When the crew tried to decrease the power to the right engine, the engine power increased instead of decreasing which is unusual. All the instruments showed a normal readings and pilots were unable to figure out the problem. Just after, the left engine quits followed by the left engine. When they tried to restart the right engine, the engine erupts in flames. A fire in the engine would spread all over the plane. There is a fire extinguisher built to extinguish engine fire, but when triggered the engine can never be re-started again. The pilots decide to pull the extinguisher of to extinguish the flame in the left engine. Suddenly, the radio goes dead result of engine failure. The right engine provides power to the radio but it spins very slowly. Pilots try to start the **Auxiliary power unit (APU)** which would provide power to the radio instruments. The APU is started. The pilots also try to restart the engine, but failed. The ATC instructs the pilots to return to the airport but, the plane lacks altitude and performing a 180 degree turn would reduce the speed which would stall the flight. The pilots try to land on the ground but, the area was covered with trees. The pilots lift nose to slow the plane and perform a soft landing and the trees would act as a cushion. Fuselage was broke into 3 pieces.

Investigation

The engineers reported that the burnt engines had number of parts missing. Later, the recovery team finds 500 fragments of the engine. This shows that the engines were coming apart during the flight. The theory suggest a **surge in the engines**, Where the engine fans are disrupted and the air flows in the opposite direction in the engine that is from the back of the engine to its front. The reason for the surge is because of damaged blades which were not efficient in directing the airflow in the engines. It is found that the blades were struck by ice. The investigation team found that there might be a possible built up of ice even after the de-icing procedure was done. When the plane took off the wings would usually curve which would have caused the ice to brake up and the broken ice could have damaged the engine blades which would have caused the surge. But, such ice damages is tolerated by the engines blades and do not necessarily cause issue. During such surges the pilots are trained to reduces the power. But, in this case when the pilot tried to reduces the power the power abruptly increased this was due to a system called as **Automatic thrust restoration (ATR)** which automatically increases the thrust while climbing this was introduced as a safety feature. The main purpose of the ATR was that it would make pilots impossible to reduce the power low a dangerous level. This worsen the engine surge.

Changes that were brought after the incident

Reports point airline for inadequate de-icing procedure and checking. The All pilots were also given training on how to act during the surge with the ATS systems on action.

Reference

1. How Did Scandinavian Flight 751 Survive This? — Mayday Air Disaster
<https://www.youtube.com/watch?v=4u7QZkcoono>